Hanlei Shi

shihanlei23@mails.ucas.ac.cn | +8618777269257

No. 1, Xiangshan Zhinong, Xihu District, Hangzhou, Zhejiang, China

Education

2026.09 - Present **University of Chinese Academy of Sciences**
Ph.D. Candidate, Computer Science and Technology (Artificial Intelligence)

2023.09 - 2026.06 **University of Chinese Academy of Sciences**
M.Sc. in Artificial Intelligence GPA: 3.8/4.0

2017.09 - 2021.06 **Xidian University**
B.Eng. in Information Countermeasure Technology GPA: 3.6/4.0

Work Experience

Beijing Hongdong Technology Co., Ltd. Jul. 2021 – Sep. 2023
Software Test Engineer

- Performed embedded software testing, automation scripting, and code analysis for reliability validation in engineering systems.

Projects

Multimodal Empathetic Dialogue System for Social-AI Interaction 2024 – Present

- Developed **talking-head generation algorithms** using **Chain-of-Thought** and **multi-stage diffusion denoising** for high-fidelity lip-synced video synthesis.
- Proposed a **semantics-driven paralinguistic motion modeling framework**, enabling fine-grained facial motion generation guided by speaker intent.
- Engineered a **real-time avatar interaction system** with WebGL optimization, achieving <50 ms latency.
- Led cross-modal alignment and **conversation-level emotion modeling**, introducing local–global fusion and emotion hotspot localization for ERC.
- Outcomes include:
  - Expressive talking-head generation: submissions to *Pattern Recognition* and *Neurocomputing*, and granted patents.
  - Multimodal emotion recognition: submission to *IEEE Transactions on Affective Computing* and granted patents.
  - Emotion recognition in conversation: international conference paper under review.

Zhejiang Vanguard Project: Digital Therapeutics for Depression Detection 2024 – Present

- Developed a visual biomarker extraction system for depression detection in low-resource settings by integrating vision models with LLM-augmented questionnaire data for enhanced visual–text synergy.
- Led multimodal fusion research and proposed the **HOPE hierarchical framework**, extending cross-modal alignment methods for more robust visual–language integration.
- Achieved **1st Place (94.51% accuracy)** in the ACM MM 2025 Multimodal Personality-aware Depression Detection Grand Challenge (Young Adult Track), outperforming the runner-up by 1.4% and the baseline by 49.14%; served as lead author of the accepted ACM MM 2025 Grand Challenge Track paper.

Publications

** Co-first author, † Advisor*

1. **Hanlei Shi**, et al. *Think-Before-Draw: Decomposing Emotion Semantics & Fine-Grained Controllable Expressive Talking Head Generation.* Pattern Recognition, Accepted, 2026. Paper Link Github Link
2. **Hanlei Shi***, et al. *HOPE: Hierarchical Fusion for Optimized and Personality-Aware Estimation of Depression.* In Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM '25) — Grand Challenge Track. DOI: 10.1145/3746027.3762063. Paper Link,
3. Xiaodong Xiang, **Hanlei Shi**†. *When words move: Inferring semantic intent to animate expressive talking heads.* Neurocomputing, Accepted, 2026. Paper Link
4. Yu Liu, Leyuan Qu†, **Hanlei Shi**, et al. *From Coarse to Nuanced: Cross-Modal Alignment of Fine-Grained Linguistic Cues and Visual Salient Regions for Dynamic Emotion Recognition.* IEEE TAFFC, Accepted, 2026. Paper Link
5. Haoxun Li, Yuqing Sun, **Hanlei Shi**, et al. *MSF-SER: Enriching Acoustic Modeling with Multi-Granularity Semantics for Speech Emotion Recognition.* Accepted by ICASSP 2026. Paper Link
6. Yu Liu* , **Hanlei Shi***, et al. *Centering Emotion Hotspots: Multimodal Local-Global Fusion and Cross-Modal Alignment for Emotion Recognition in Conversations.* Under Review, 2026. arXiv Link
7. Taihao Li†, Leyuan Qu†, **Hanlei Shi**, et al. *Facial Component-Based Expression Editing Method, Electronic Device, and Readable Storage Medium.* Chinese Patent CN119152082B.
8. Taihao Li†, Leyuan Qu†, Yu Liu, **Hanlei Shi**, et al. *A Dynamic Expression Recognition Method, Electronic Device, and Computer-Readable Storage Medium.* Chinese Patent CN119206837B.
9. Taihao Li†, Leyuan Qu†, Haoxun Li, Jiaxi Hu, **Hanlei Shi**. *A Method for Emotional Speech Synthesis with Controllable Fine-Grained Emphasis.* Chinese Patent CN 119207366B.

Skills

Core Research Skills: Multimodal affective computing, generative modeling (diffusion models, GAN), multimodal alignment(CLIP, Video-LLaVA), LLM/VLM fine-tuning (LoRA, adapter tuning), Chain-of-Thought reasoning, prompt engineering.

Programming & Frameworks: Python, PyTorch, HuggingFace Transformers, C, matlab.

Engineering & Deployment: Real-time avatar systems, API integration, scalable inference.

Honors & Awards

UCAS Outstanding Graduate, Hangzhou Institute for Advanced Study, UCAS, 2026

Graduate National Scholarship, Hangzhou Institute for Advanced Study, UCAS, 2025

Merit Student, Hangzhou Institute for Advanced Study, UCAS, 2025

First-Class Scholarship, Hangzhou Institute for Advanced Study, UCAS, 2024 – 2025

Second-Class Scholarship, Hangzhou Institute for Advanced Study, UCAS, 2023 – 2024

Second-Class Scholarship, Xidian University, 2020–2021

Second Prize, Shaanxi Division, National Undergraduate Mathematical Modeling Competition, 2019

First Prize, CAMCA International Mathematical Modeling Competition, 2018